

Article

Design, Construction, and Flight Performance of an Electrically Operated Fixed-Wing UAV

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Abstract: The development of unmanned aerial vehicles (UAVs) has attracted much attention in the global community and aviation industry. As UAVs have the potential to be applied for multiple missions, the level of research into improving their design and flight performance has also increased. In this context, the present paper aims to present the design, construction, and flight performance of an electrically operated fixed-wing UAV. As a first step in the design process, key performance requirements are defined, such as the thrust required, the stall speed, the minimum drag velocity, and the minimum power velocity. Wing and associated power loadings are calculated according to the defined performance requirements. In addition, payload and endurance requirements are set up in order to determine the wing and tail areas, the total mass, the power requirements, and the motor size. Aerodynamics and stability designs are also calculated. After the completion of the design process, the manufacturing of the UAV follows by using appropriate materials. Flight tests were carried out for the evaluation of the UAV's flight performance, where the success of the design was demonstrated.

Keywords: unmanned aerial vehicle; electrically operated; aerodynamic design; stability analysis; manufacturing; flight tests; flight performance



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1. Introduction

The development of unmanned aerial vehicles (UAVs) has attracted much attention in the aviation industry for decades. For many years, the UAV market was predominantly driven by military requirements like reconnaissance, search, and rescue [1]. On the other hand, the most dominant factors for civilian UAV applications are focused on the operational costs, safety, and multirole capabilities [2]. Additionally, low-cost UAVs have a significant influence on the overall value and specific applications in the aviation industry [3].

The design requirements to evaluate and configure the development of UAVs are of utmost importance. In this context, aerodynamic design plays a crucial role in the UAV's conceptual design, as it strongly influences the UAV's structure and systems, and therefore, its construction process. Additionally, a well-defined aerodynamic design enables a UAV to have a good flight performance, whereas operational costs are reduced and energy consumption becomes lower. At the beginning of the UAV's conceptual design, many parameters can be determined [4].

A wide number of research studies concerning the design and flight testing of small scale fixed-wing UAVs exist in the international literature [5,6]. Chung et al. [7] proposed an aerodynamic performance design analysis for an experimental flying wing UAV, based on the matching plot, weight estimation approach, and conventional aircraft design, whereas Karageorgiou et al. [8] presented a step-by-step analysis of the critical design process for a small-scale fixed-wing UAV, being able to transport blood bags to remote locations.

Additionally, high-energy-density systems within UAVs, with lower weights and better performance under the same flight conditions, are a research challenge [9]. Adawy et al. [10] presented a robust design procedure for the designing, manufacturing, and testing of an electrically powered UAV for carrying the highest payload possible, whereas Sarvesh et al. [11] proposed a holistic approach for designing, prototyping, and testing an electric-powered fixed-wing hybrid VTOL UAV. Moreover, the weight strongly influences the UAV's energy consumption, and thus, it makes the design of the battery a challenging process, as it aims to satisfy the UAV's required range and endurance [12]. In this context, Traub [13] proposed a synthesized UAV mathematical design model suitable for sizing and configuration design of small solar-powered and battery-powered electric systems to enhance the final UAV design characteristics and performance for high-endurance missions.

Moreover, due to the diversity and complexity of UAV systems, their reliability is becoming a prominent issue demanding up-to-date solutions tailored to the UAV design requirements and specifications. In this context, Ahmed and Jenihhin [14] proposed a cross-layer reliability model tailored to UAVs' onboard intelligence, whereas Song et al. [15] presented a cascade ensemble learning approach for the multi-level reliability evaluation of an aeroengine turbine rotor system.

In light of the above information, although there are a wide number of studies concerning the design of electrically operated small-scale, fixed-wing UAVs, and only a few of them use low-fidelity software and in-house tools. In this context, the main purpose of the present study lies in the design of an electrically operated fixed-wing UAV, by addressing a basic traditional design approach and also taking into consideration the characteristics and constraints of flights for small-size, fixed-wing UAVs. In addition, this paper presents the construction process and flight performance testing of the above UAV for low-altitude and straight-level flight.

Therefore, the principal contributions of this work are summarized in the following points:

- Addresses a basic traditional design approach by also taking into consideration the characteristics and constraints of flights for small-size, fixed-wing UAVs;
- Uses low-fidelity software and in-house tools to support the design and performance of the fixed-wing UAV, by also keeping low the required propulsion;
- Addresses low construction costs and advanced flight autonomy focusing on the fail-safe flight operation of the developed UAV.

In total, the work in this study is divided into three main parts. The first part relies on a step-by-step presentation of the critical design process including the conceptual design with the mission and performance requirements, the preliminary design of the most critical performance specifications, the endurance determination, the take-off distance determination, and the wing and tail design. The second part presents the fabrication methods used during manufacturing, whereas the third part presents the ground-testing and checking if these are valid-testing results upon which the final conclusions are drawn.

In general, the overall design methodology is mainly related to the definition of the performance requirements for the fixed-wing UAV such as the stall speed, the minimum drag and minimum power requirements, the thrust requirements, as well as the selection of the wing/tail airfoils. In more detail, the wing area and power requirements are calculated based on the UAV's weight and the appropriate wing and power loadings. Therefore, the UAVs electric motor specification is defined with regards to the electric power budget. Furthermore, airfoil selections, aerodynamic design, and stability analysis are considered, and these parameters are associated with the UAV's wing area. Additionally, the manufacturing procedure with respect to the fuselage, main wing, left/right wingtips, and tail sections is reported for the designed UAV, where appropriate materials are used to build the UAV. Additionally, two flight-performance tests are presented, where the first test observes the basic flight motion of the UAV with a remote controller, and the second test uses the flight control system and built-in gyroscope and accelerometer to record the flight parameters of the developed UAV.

2. Design Process

This section includes the overall design analysis of a low-speed and low-altitude fixed-wing UAV.

2.1. Conceptual Design

In practice, most designers often essentially complete a baseline configuration of the UAV in its first design process. Then, several iterations of the baseline configuration need to be carried out. The number of iterations depends on the designers' experience and the design database.

In addition, as one of the main challenges of the present study is to reduce the production costs, while achieving design goals, the optimum and efficient use of the propulsion system is of utmost importance. By taking into consideration the interaction between the lift and drag forces, there is a specific flight speed that minimizes the engine power required for a particular cruise altitude. In general, the aim is to reduce the required engine power through optimal design, and thus, the overall weight of the UAV can be adjusted accordingly.

Therefore, the conceptual design process begins with an evaluation of the mission and performance requirements to be achieved, by considering the regulations set by [16]. The most critical mission requirements for the developed UAV are presented below:

- Take-off distance limited to 30 m of runway;
- Maximum wing projected area not exceeding 0.70 m²;
- Eco-friendly performance with the use of electric power motors;
- Propulsion system, electronic parts and payload had to fit in a transportation box with dimensions 1100 mm × 100 mm × 175 mm;
- Take-off weight of 6 kg (payload 1 kg plus net weight estimation 5 kg);
- Cruise flight speed had to range between 13 m/s and 16 m/s;
- Cruise height should not exceed 500 m;
- Stall speed should be at about 11.5 m/s;
- Endurance should be at least 1.5 h.

Additionally, by keeping the required propulsion of the UAV for achieving a horizontal straight flight low, the UAV's flight autonomy increases. Moreover, by selecting the appropriate wing and tail fin, the stability of the UAV can be maximized, as well as the overall UAV's flight performance. In this context, the selection of the wing and tail airfoils is divided into three main stages:

- Theoretical estimation of the required lift coefficients;
- Analysis of the candidate airfoils, by using a low-fidelity tool, the well-established XFLR5 software;
- Selection of the most suitable airfoils for the wing and tail.

Finally, all of the above performance requirements should be defined, giving emphasis on flight autonomy and low construction costs, while meeting the initial mission specifications.

2.2. Preliminary Design and Performance Requirements

2.2.1. Stall Speed

The stall speed is a limit to the minimum speed at which a UAV can safely lift off from the ground. In the classical lift–weight equation, the left and right-hand sides are the weight W and lift L , respectively. Due to the balance of forces, the wing area S can be moved to the left side. Given the stall speed v_s and the maximum lift coefficient C_{Lmax} , the wing loading (W/S) is calculated as

$$\frac{W}{S} = \frac{1}{2} \rho_0 v_s^2 C_{Lmax} \quad (1)$$

where S is the wing area and ρ_0 is the air density at the sea level.

2.2.2. Minimum Drag and Minimum Power Velocities

The minimum drag and minimum power velocities V_{md} and V_{mp} , respectively, play a key role in the UAV's flight performance estimation. For achieving the UAV maximum flight autonomy, usually an intermediate choice of the above speeds V_{mp} and V_{md} is used, but at the same time, the minimum drag and the minimum power consumption should be considered. The only obstacle for the calculation of the above speeds V_{mp} and V_{md} is the initial specifications of the UAV design, i.e., the interval should be in the range of 13–16 m/s, as mentioned previously in Section 2.1. V_{md} can be expressed with the following formula:

$$V_{md} = \left[\frac{4}{\pi \cdot e \cdot AR} \cdot \frac{W^2}{S} \cdot \frac{1}{\rho^2} \cdot \frac{1}{C_{D0}} \right]^{0.25} \quad (2)$$

where $\pi = 3.14$, e is the Oswald coefficient, AR the aspect ratio, ρ the air density at a specific flight altitude and C_{D0} the zero-drag coefficient. Raymer [17] proposes the following empirical formula for the Oswald coefficient e for straight wings:

$$e = 1.78 \cdot \left(1 - 0.045 \cdot AR^{0.68} \right) - 0.64 \quad (3)$$

2.2.3. Thrust Required

The calculation of the required thrust in steady flight situations affects the overall flight autonomy performance of the UAV. In general, during the cruising phase, the four main forces existing on a UAV are balanced, so thrust T is equal to the total drag D of the UAV.

Total drag D is a function of the aerodynamic coefficient C_D , which is a function of parasite drag C_{D0} and induced drag C_{Di} , as follows:

$$C_D = C_{D0} + C_{Di} \quad (4)$$

where

$$C_{Di} = K \cdot C_L^2 \quad (5)$$

and

$$K = \frac{1}{e \cdot \pi \cdot AR} \quad (6)$$

Substituting Equations (5) and (6) into Equation (4), the general parabolic polar drag curve of the UAV can be formulated as follows:

$$C_D = C_{D0} + K \cdot C_L^2 = C_{D0} + \frac{C_L^2}{e \cdot \pi \cdot AR} \quad (7)$$

With the help of Equation (7), the total drag D of the UAV is given by the following expression:

$$\frac{1}{2} \cdot \rho \cdot V^2 \cdot S \cdot C_D \rightarrow D = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S \cdot \left(C_{D0} + K \cdot C_L^2 \right) \quad (8)$$

where V is the airspeed of the UAV at a specific flight altitude where the density is ρ . As $L = W$ in the cruising phase, the lift force coefficient C_L can be obtained as follows:

$$C_L = \frac{2 \cdot W}{\rho \cdot V^2 \cdot S} \quad (9)$$

In this case, the total drag of the UAV can be calculated as follows:

$$D = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S \cdot C_{D0} + \frac{2 \cdot K}{\rho \cdot V^2} \cdot \left(\frac{W}{S} \right)^2 \quad (10)$$

The first term of Equation (10) indicates that drag D (or thrust T) increases with the square of the airspeed V . This term is related to the parasite drag of the UAV, which increases with the increase in the dynamic pressure $1/2\rho V^2$. On the other hand, the second term of Equation (10) represents the induced drag of the UAV, which decreases as the speed V increases. This term also includes the effect of the C_L on the drag D . The higher the speed V of the UAV is, the lower the value of C_L .

In general, by reducing the speed V of the UAV during the cruising phase, C_L must be increased so that the lift L equals the weight W of the UAV. Therefore, with the decrease in the airspeed V , the second term of Equation (10) increases exponentially. On the other hand, by increasing the airspeed V , the C_L will decrease, so in this way, both terms of Equation (10) benefit, with the only conclusion being that the increase in the flight speed V and the choice of a relatively small angle of attack (small C_L) will bring autonomy advantages as the total drag D will be reduced. In this context, advantages of required thrust are taken, while at the same time, the UAV can cover a greater distance for a given time period.

2.2.4. Selection of the Wing/Tail Airfoils

The selection of the appropriate airfoils (wing, tail) will be made according to the maximum autonomy performance of the UAV for the given specifications. For the simplicity of the whole design process, the choices of the wing/tail airfoils are limited to 4-digit NACA airfoils [18].

With respect to the main wing airfoil selection of the UAV, one choice could be the NACA 0006 airfoil, which is a relatively symmetrical thin airfoil with uniform pressure distribution. This airfoil has a low drag coefficient C_D , which makes it more suitable for the tail section of the UAV. Also, there are airfoils such as the NACA 4424, which is a relatively thicker airfoil that offers a large lift coefficient C_L but at the same time it also gives a large drag coefficient C_D . This characteristic makes it efficient at low-subsonic speeds (due to the increased C_D) and useful for UAVs that fly at these speeds and need a lot of lift (due to the increased C_L).

After some initial calculations, the wing airfoil that combines all the above characteristics is the NACA 4412. The aerodynamic characteristics of the NACA 4412 airfoil also give a wing area of 0.49 m^2 and a stall speed of 11.7 m/s . The exact determination of the aerodynamic coefficients C_L , C_d , and C_m of the selected wing airfoil is presented in Figure 1, which were made with the help of the XFLR 5 software tool [19], based on the basic geometrical characteristics of the UAV.

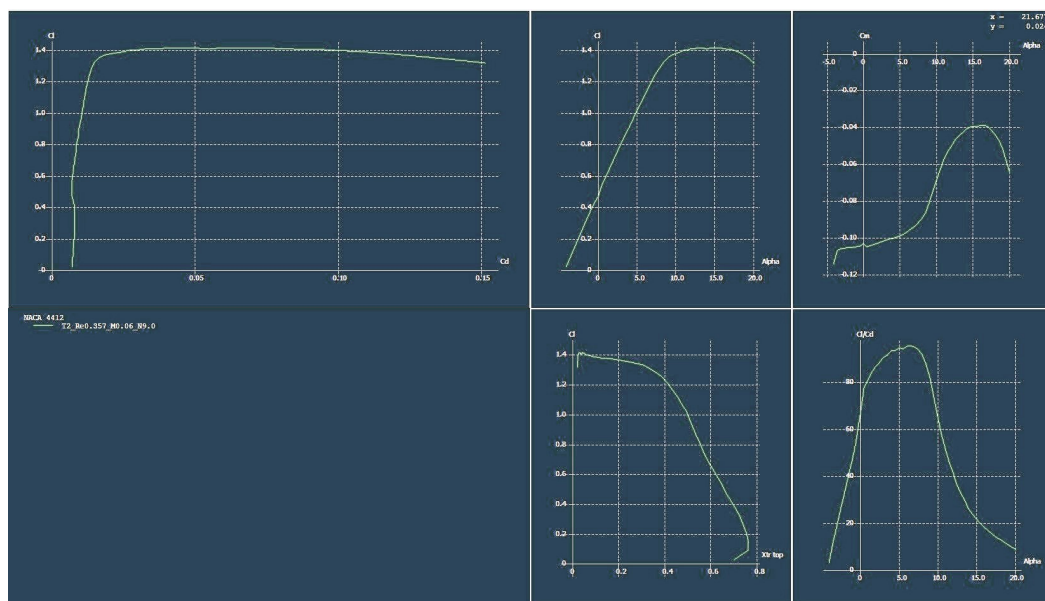


Figure 1. Determination of the basic aerodynamic coefficients for the wing airfoil NACA 4412.

With regards to the tail airfoil selection of the UAV, a symmetrical airfoil is usually used, giving the advantage of uniform pressure distribution, and consequently, the same absolute value of C_L for every equal and opposite angle of attack α . Also, in a symmetrical airfoil, its center of pressure does not change for each angle of attack α as it does in the case of an asymmetrical airfoil. Furthermore, the tail airfoil characteristics determine the value of the parasite drag C_{D0} . In general, the thinner the tail airfoil is, the lower the value of the parasite drag C_{D0} is, and that means, the value of the minimum drag velocity V_{md} is greater.

In this context, after some initial calculations, NACA 0009 is the most appropriate tail airfoil for the developed UAV. Of course, further analysis on the selected tail airfoil and its main aerodynamic coefficients C_L , C_d , and C_m , was carried out with the help of XFLR5 software, as presented in Figure 2, based on the basic geometrical characteristics of the developed UAV.

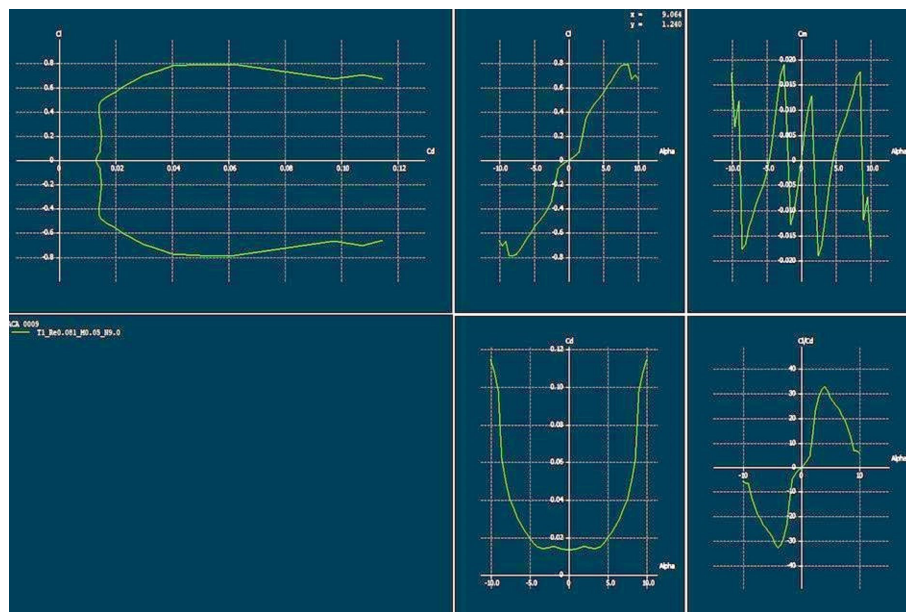


Figure 2. Determination of the basic aerodynamic coefficients for the tail airfoil NACA 0009.

3. Endurance Determination

3.1. Aspect Ratio Calculation

According to Equation (2), all the information such as the selected wing loading, the cruise flight height (or equivalent the density ρ), and the wing airfoil aerodynamic characteristics is known, and the only factors that will change over time is the aspect ratio AR and the Oswald coefficient e , as it depends on the AR value. Initially, three typical AR values (8, 10, 12) are chosen. AR selection is examined in terms of the minimum drag velocity V_{md} and minimum power velocity V_{mp} . In case the mean value of V_{md} and V_{mp} for the aforementioned AR values (8, 10, 12) is out of the original specifications (see Section 2.1), the next step will be to set up modifications on the wing loading or the wing airfoil aerodynamic characteristics.

Taking the first value ($AR = 8$), it is shown that due to the relatively small wingspan b , benefits are provided at relatively higher speeds (due to increased V_{md}), compared to the other AR values (10, 12), but in the end, this value is rejected because at the same time V_{mp} increases, the energy consumption consequently increases. On the other hand, by taking the third value ($AR = 12$), due to the larger wingspan b , the drag increases and a lower V_{md} is also observed. It can also be noticed that large wingspans are suitable for conditions of minimum energy consumption, due to the smaller V_{mp} . Finally, this option is rejected because the only advantage compared to the second value ($AR = 10$) is the lower minimum drag velocity by 0.33 m/s, but at the same time, due to the size of the wing,

there is a high possibility of creating problems of structural strength, and consequently, the overall construction will require more weight. Thus, the second value ($AR = 10$) will be chosen, as in general it combines the advantages of the other two AR values, due to the fact that it has the appropriate spread for V_{md} conditions and, at the same time, keeps the V_{mp} low, offering advantages of autonomy, without any particular risk of creating structural problems.

Next, as the aspect ratio AR has been chosen by taking the value 10, the wing span b can be calculated through the following formula:

$$AR = \frac{b^2}{s} \rightarrow b = \sqrt{AR \cdot s} \rightarrow b = 2.21 \text{ m} \quad (11)$$

The next parameter that needs to be clarified is the taper ratio λ , where C_t is the airfoil chord at the wing tip and C_r the airfoil chord at the wing root:

$$\lambda = \frac{C_t}{C_r} \quad (12)$$

With λ known, the next step is to find the aforementioned airfoil chords to give a detailed picture of the main wing. By combining the following two formulas for the mean aerodynamic chord $\bar{c}$:

$$\bar{c} = \frac{2}{3} \cdot c_r \cdot \left(\frac{1 + \lambda + \lambda^2}{1 + \lambda} \right) \quad (13)$$

$$\bar{c} = \frac{b}{AR} \quad (14)$$

the final values for $\bar{c}$, c_t , and c_r are obtained:

$$\bar{c} = 0.22 \text{ m } c_r = 0.27 \text{ m and } c_t = 0.17 \text{ m} \quad (15)$$

3.2. Angle of Attack Calculation

After the calculation of the aspect ratio AR , the next step is to find the most appropriate angle of attack α that the UAV can fly during the cruise phase. Typical values for the angle of attack α of small-scale, fixed-wing UAVs are 2° to 4° . First of all, flight speeds should be calculated for the aforementioned angles of attack. Secondly, these flight speeds are examined in terms of the minimum drag velocity V_{md} and minimum power velocity V_{mp} . Finally, for each angle of attack, the thrust required to achieve the relative cruise flight speed was found.

According to the calculation results, the first case ($\alpha = 2^\circ$) is rejected because the flight speed takes the value of almost 17 m/s, which is outside of the original specifications (see Section 2.1), whereas at the same moment it does not coincide with any intermediate value of minimum drag and minimum power velocities. That means that the UAV will fly outside of the predefined conditions. On the other hand, for the next two cases ($\alpha = 3^\circ$ and $\alpha = 4^\circ$), the flight speeds are within the original specifications (almost 16 m/s and 15 m/s, respectively) and also coincide with an intermediate value of V_{md} and V_{mp} .

The above statements clearly show that the appropriate angle of attack for maximizing the endurance of the UAV, during the cruising phase, is $\alpha = 3^\circ$ as it requires almost 0.04 kg less thrust than the other case ($\alpha = 4^\circ$), and at the same time, the UAV will fly at about 0.94 m/s faster, thus covering more distance in a given time, using less energy consumption. This naturally occurs because as C_L increases, C_D usually increases, and as a lower speed for the cruise phase is required due to a large C_L , the result is to increase the induced drag C_{Di} .

3.3. Engine Selection

According to the restrictions presented in Section 2.1, the developed UAV requires a flight time of at least 1.5 h and should be eco-friendly. To achieve these characteristics,

an electric motor is selected. This choice is enhanced by the fact that, in general, electric motors have many benefits, like low cost, high thrust-to-weight ratio, satisfactory consumption, ease of installation, simple operation, long lifetime, minimal maintenance, and zero operational costs [20].

Regarding the number of electric motors to be installed on the UAV, the following factors should be considered:

- Safety of the flight in the event of a breakdown;
- Achievement of required thrust;
- Complexity;
- Weight.

Next, as a thrust of less than one kilogram is achieved very easily with the use of a single (suitable) electric motor, the use of a second or more electric motors is deemed unnecessary. Due to its simplicity, the reliability of an electric motor is quite high, and thus, there is no need to use a second or more electric motors. Additionally, the use of more than one electric engine increases the complexity of construction and installation as well as the weight. Based on the above information, the developed UAV will have one electric motor.

Additionally, the LiPo-type battery is chosen due to lower internal resistance and higher energy-to-weight ratio, and thus, the developed UAV is safer. For the present study, the Sunnysky 2820 800 kV electric motor is chosen [21]. This model can collaborate with various propellers and batteries, but in the case of the developed UAV, where one of the main design factors is autonomy, the case with the highest efficiency ratio (g/W) is chosen, i.e., for battery LiPo 3s and 330 × 1650 mm propeller. In accordance with the engine performance data, the estimated power consumption is approximately 70 W. With an 11.1 V battery, this translates to 6.3 A/h. Consequently, the developed UAV is estimated to achieve a 1 h autonomy. To fulfill the initial autonomy target of 1.5 h, a minimum of an 11.1 V 9.5 A Lipo battery is required, which weighs approximately 0.8 kg.

An additional advantage of the selected motor model is its compatibility with the battery that possesses the minimum number of cells required for optimal motor operation. This not only minimizes weight but also reduces costs and overall size. It is noteworthy that in practical scenarios, it is customary to employ a battery with a capacity at least 30% larger than the calculated requirement. This precautionary measure accounts for potential failures and ensures avoidance of critical battery discharge.

4. Take-Off Distance Determination

In general, the take-off performance plays a significant role in the design process of a fixed-wing UAV. It affects the take-off distance, the take-off speed, the ground it can operate, its geometry, the complexity of design and construction, as well as the UAV's cruising performance. The preliminary design requirements in Section 2.1 set the take-off distance at less than 30 m. The take-off distance can be derived by the following expression [16]:

$$x = -\frac{1}{2B} \log_e \left(\frac{A - BV_{TO}^2}{A} \right) \quad (16)$$

where the factors A and B have constant values for a UAV with specific characteristics and V_{TO} is the take-off velocity. In more detail, factor A depends mainly on the quality of the runway (friction coefficient μ) and the static thrust T_0 of the engine during the take-off phase, whereas factor B depends on the aerodynamic characteristics of the developed UAV. Factors A and B can be expressed by the following formulas:

$$A = g \left(\frac{T_0}{W} - \mu \right) \quad (17)$$

$$B = \frac{g}{W} \left(\frac{1}{2} \rho S (C_D - \mu C_L) + a \right) \quad (18)$$

Another important factor in the determination of the take-off phase is the take-off velocity V_{TO} , which is mainly affected by the take-off angle of attack α . In more detail, according to Figure 3, the wing's stall, based on the diagram C_L - α , starts from an angle of attack $\alpha \approx 17^\circ$. Additionally, from the angle of attack $\alpha \approx 13^\circ$, the lift coefficient C_L remains the same, but C_D increases (diagram C_D - α). In short, an angle of attack that can offer a large C_L without a particularly large C_D , will have tolerances in case of mishandling and the take-off speed will always be higher than the stall speed V_s , Equation (1). Eventually, after extensive analysis in our study, it was found that the developed UAV, without the modification of the initial characteristics (see Section 2.1), could not take off in less than 30 m, so the addition of flaps was chosen.

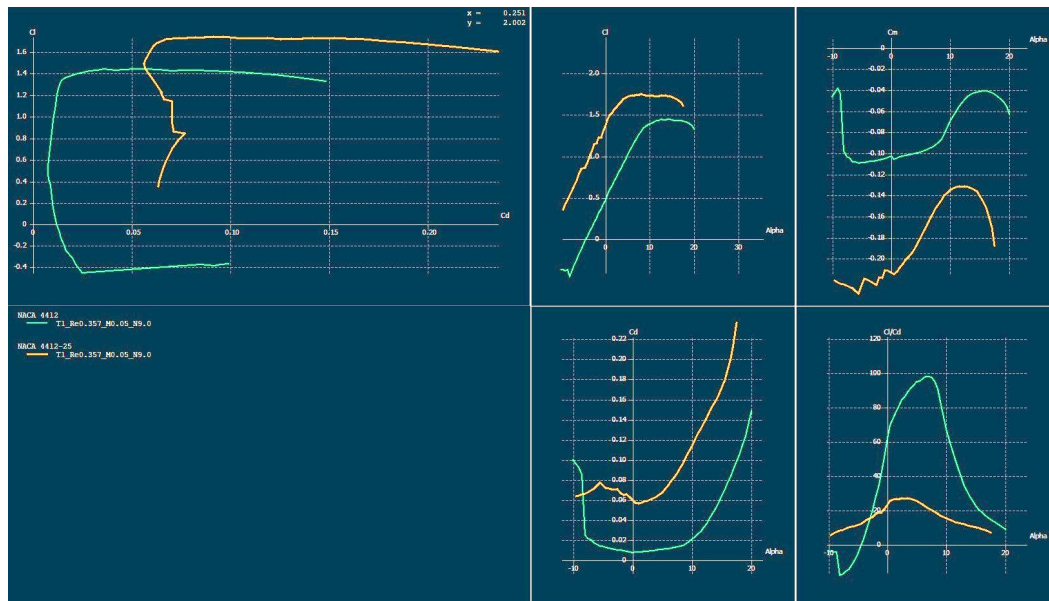


Figure 3. Variations in the aerodynamic coefficients for the NACA 4412 wing airfoil; (a) without a flap (green lines) and (b) with a plain flap deviated by 25° (orange lines).

Following the above process, flaperons are selected for better aerodynamic performance, by introducing a simple flap (plain flap) due to ease of design and ease of construction (thus minimizing the total construction time of the developed UAV), the reduced construction weight, and the ease of replacement in case of failure (due to the simplicity of construction and installation), while at the same time can offer satisfactory results. In the case of plain flaps, C_{Lmax} reaches the value of 2.4, while at the same time, the optimal length of the flap is about 30% of the airfoil and the optimal angle of deviation is up to 60° [22].

5. Tail Design

In this section, the analysis of the UAV tail is performed, which is a quite important and decisive step, due to its effect on the flight behavior and the endurance performance of the designed UAV. In the present study, an A-tail configuration (otherwise known as an inverted V-tail configuration) for the developed UAV is chosen [23]. This tail configuration combines the characteristics of horizontal and vertical tail fins and at the same time offers the following advantages:

- Reduced tail drag and reduced yaw side movement in case the electric engine is placed at the rear of the fuselage;
- Easy, cheap, and light construction;
- Zero tail–fuselage interaction.

Once the wing geometry is determined, the analysis starts by randomly selecting the geometric characteristics of the tail, in order to combine both stability and autonomy performance for the overall UAV. The basic geometric characteristics of the tail section for the developed UAV (see Figure 4) are presented below:

- $L = 0.67$ m;
- $h = 0.34$ m;
- $a = 0.47$ m;
- $b = 0.1$ m.

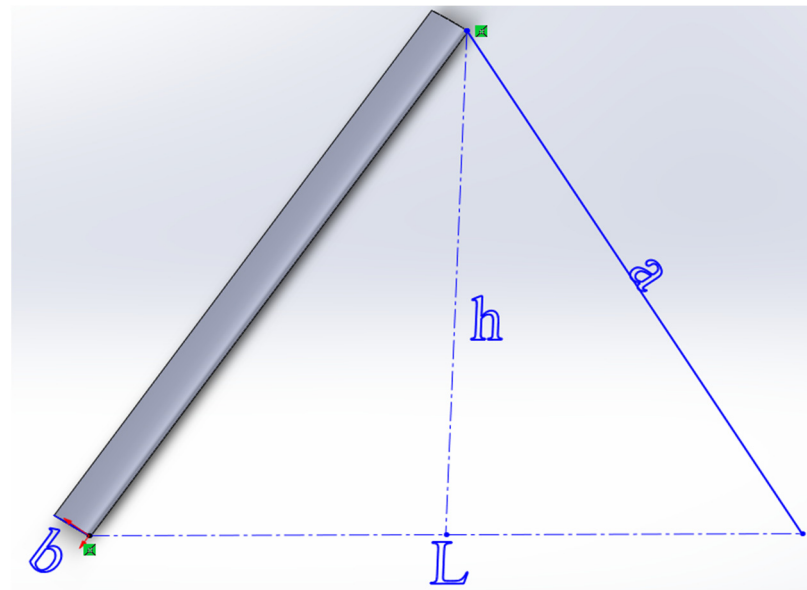


Figure 4. Characteristic dimensions of the tail for the developed UAV.

At this stage, the appropriate tail airfoil must also determine the appropriate distance between the center of gravity (C.G.) and the center of pressure for the tail fin (see Figure 5). C.G. was selected in the rear extreme position to increase the required V_{md} for better endurance performance. Additionally, the A-tail angle was chosen at about 60° .

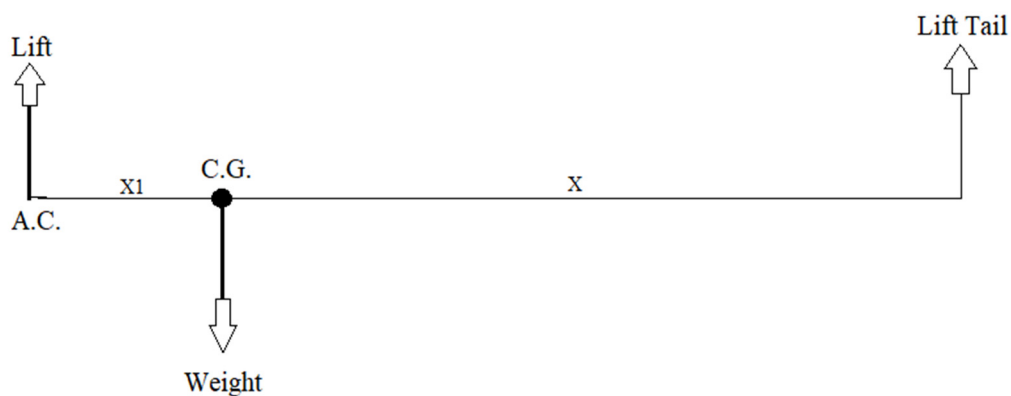


Figure 5. Critical distances between the center of gravity (C.G.) and the center of pressure in the tail fin.

Regarding the static stability of the UAV, an important factor is the static margin s_m , which is given by the following formula:

$$s_m = \frac{N.P. - C.G.}{MAC} > 0 \quad (19)$$

s_m must always be greater than zero; that means that the neutral point (N.P.) must be behind the C.G. In general, the higher the value of s_m , the greater the static stability of the UAV [24]. By substituting the corresponding values, as mentioned previously, s_m reaches out the value of 10%.

Further stability analysis follows, based on the XFLR 5 software tool. The dynamic stability of the wing–tail system was confirmed in both the lateral (see Figure 6) and longitudinal (see Figure 7) axes.

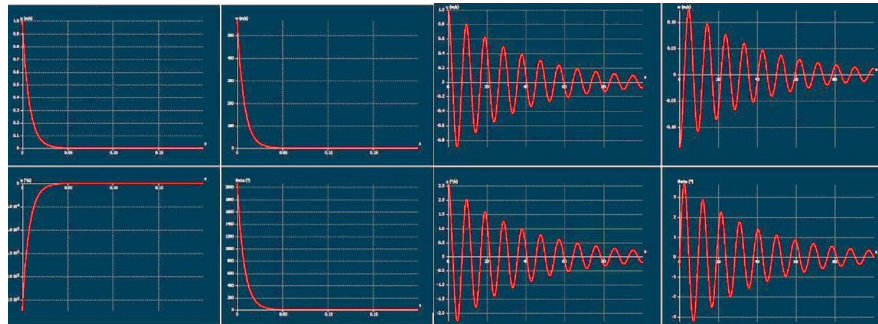


Figure 6. Dynamic stability on the lateral axis of the designed UAV.

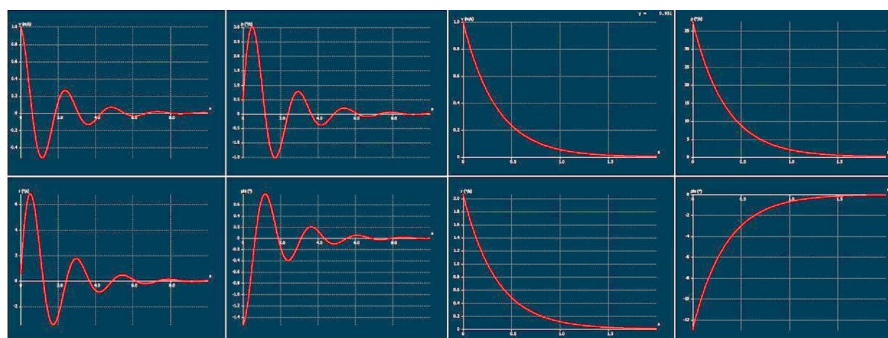


Figure 7. Dynamic stability on the longitudinal axis of the designed UAV.

Based on the aforementioned geometric data of the wing–tail system, the required thrust can be calculated in order to have a clearer picture of energy consumption. This can be carried out by adding the results for the three drags (wing, elevator section of the tail, rudder section of the tail), which provide a total thrust of almost 0.6 kg.

Finally, as the above data are known, further analysis can be carried out for the static tests of the electric motor. The results showed that for the production of a thrust at about 0.6 kg, the electric motor consumes about 60 W with a battery of 11.1 V, so as a result, it has in its current consumption $60 \text{ W}/11.1 \text{ V} = 5.4 \text{ A/h}$. In this context, in order to achieve an autonomy of at least 1.5 h, the presence of a Li-Po 11.1 V and 8.1 A battery is needed, weighing approximately 0.6 kg. Therefore, the total payload is 4.4 kg for the construction of the developed UAV.

6. Winglets Design

Winglets, in general, are used to improve the efficiency of the wing (or propeller) by reducing the induced drag due to the pressure difference between the top and bottom of the wing [25]. Winglets can also improve, in some cases, the handling characteristics of a UAV, whereas they can also be used as a vertical stabilizer. In general, they can increase the effectiveness of the AR without significantly increasing the wingspan, while also offering benefits such as reducing wing load, thus minimizing the structural requirements of the wing [26]. However, at the same time, winglets may increase the parasite drag and, thus, the complexity of the UAV's design and construction.

In general, the formulation of the developed UAV is based on the blended winglet type in order to reduce time analysis and construction complexity. Its features selection was carried out through the XFLR 5 software tool, by taking different values of dihedral angle (from 40° to 90°) and comparing results with the use of the previously selected airfoils NACA 4412 and NACA 0009 for the wing and tail, respectively. The best performance was found for the dihedral angle 45° , wingspan 0.1 m, 0.07 m offset, $+1.25^\circ$ twist angle, and NACA 0009 airfoil (see Figure 8). The results of the static stability analysis are shown in Figure 9, where it can be noticed that their addition has affected the static stability of the lateral axis of the UAV, due to the fact that the angle of attack which offers a zero pitching moment coefficient takes the value of 2.8° .

By adding the winglets into the main wing, the cruising flight speed now takes the value of 14.64 m/s, which is between the two characteristic flight speeds $V_{mp} = 12.71$ m/s (for minimum power) and $V_{md} = 16.73$ m/s (for minimum drag), whereas the lift-to-drag ratio takes the value of 25.5. The relative values for the cruising flight speed and the lift-to-drag ratio, without the consideration of winglets, are 15.86 m/s and 23.9, respectively. Another feature that is improving, when winglets are added, is the wing loading, which takes the value of 11.6 kg/m^2 . This value is almost 10% smaller than the corresponding value of 12.3 kg/m^2 without the consideration of winglets. This allows less structural strength of the main wing, and thus, less weight. Finally, it is worth noting that the winglets' addition affects the dynamic stability of the UAV in the longitudinal axis, but at legitimate levels.

The analysis when adding winglets into the main wing was completed with the final examination of the UAV's autonomy performance, based on the resulting geometric and aerodynamic characteristics. By following the same analysis as the one in Section 5, the required thrust with the addition of winglets ($T = 0.56 \text{ kg}$) is 9% smaller than the corresponding value ($T = 0.61 \text{ kg}$) without the winglets.

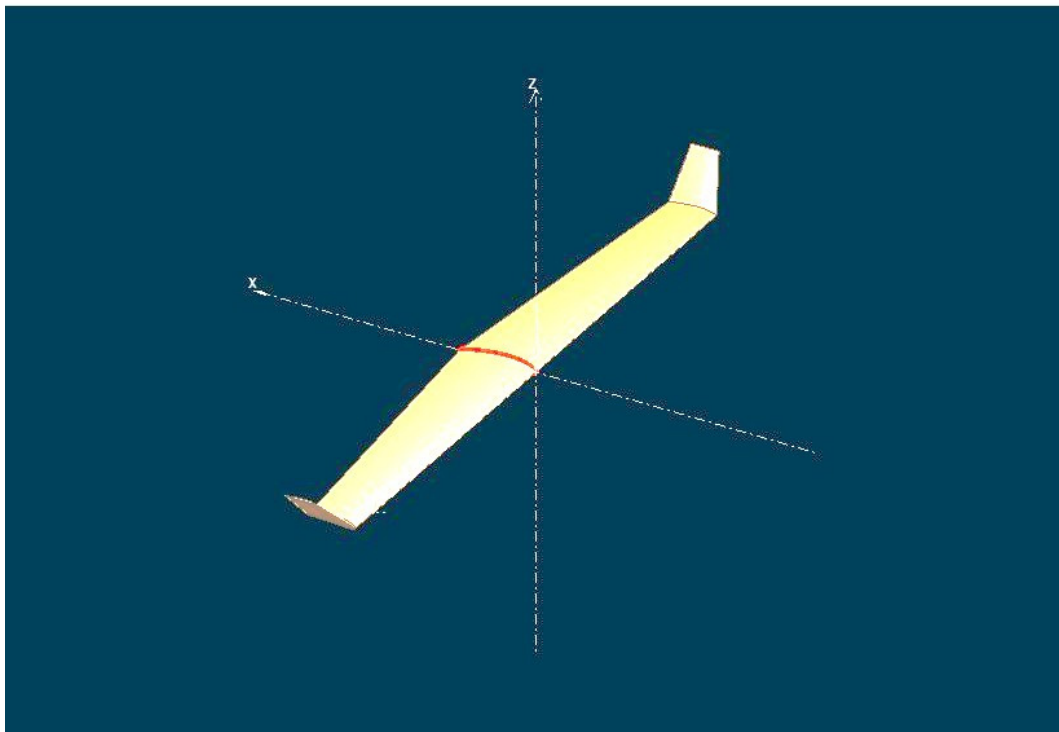


Figure 8. Winglets into the main wing of the designed UAV.

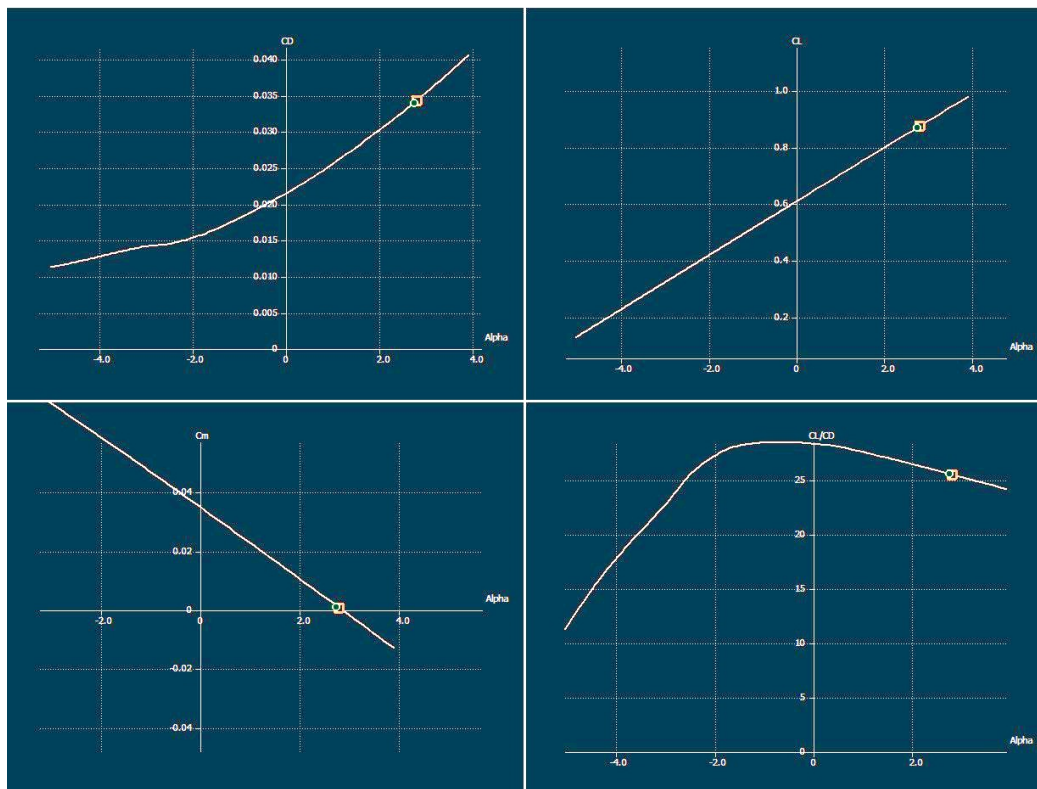


Figure 9. Static stability testing when adding the winglets into the main wing of the designed UAV.

7. Overall Design Analysis

One of the most important factors is the aerodynamic design of the developed UAV parts and the points of contact between them, minimizing the drag, thus increasing the endurance performance (see Figure 10). As the aerodynamics of the UAV are improved and confirmed, various aerodynamic simulations were performed through the SolidWorks CFD environment (see Figure 11), which ultimately showed that the UAV would be cruising at a speed of almost 16 m/s, which is much higher than the theory, but it makes sense to consider the additional drag of an entire UAV.

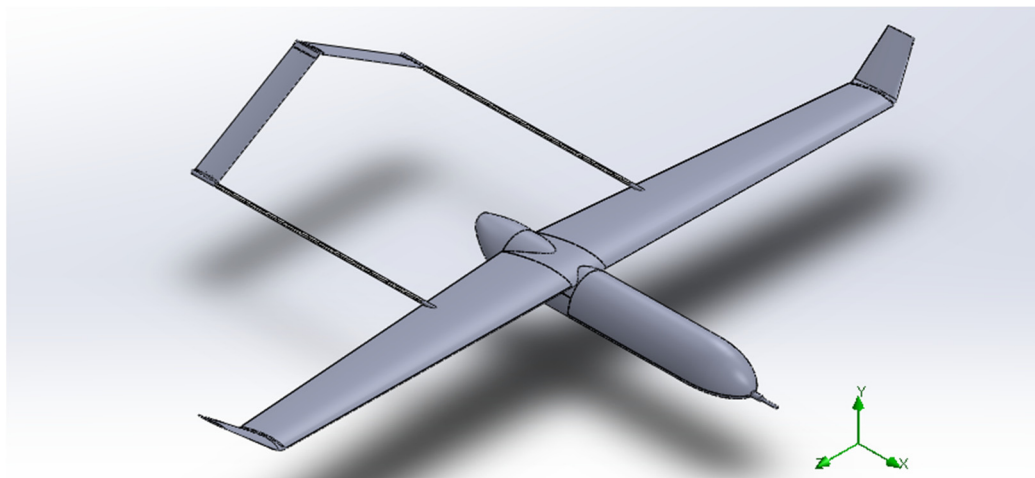


Figure 10. Developed UAV as designed in the SolidWorks software tool (version 2018).

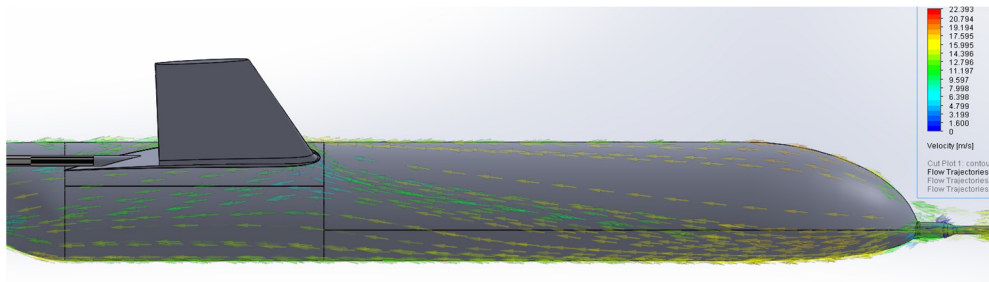


Figure 11. Aerodynamic simulation by using the SolidWorks CFD environment.

Following the results from the CFD analysis, the fuselage was re-designed to become longer and narrower and reduce inductive drag and drag friction effects. Additionally, the nose was re-designed to disperse the flow uniformly along the fuselage, whereas the “rear part” was re-designed in such a way by minimizing the turbulence flow production, having at the same time sufficient space for the installation of the engine. What can be noticed in Figure 11 is that at t , the “top” of the nose, there is a disturbance due to fuselage geometry, which changes the airflow velocity, and consequently adds surface friction and drag.

The next step is the “real” design phase, i.e., how the UAV could be designed in real time (see Figure 12). This process needs to take into consideration ergonomics characteristics, which means that any part of the developed UAV should be easily disassembled and tied, and easily maintained, whereas it must be removable for transportation purposes and has excess space for further upgrades [27].

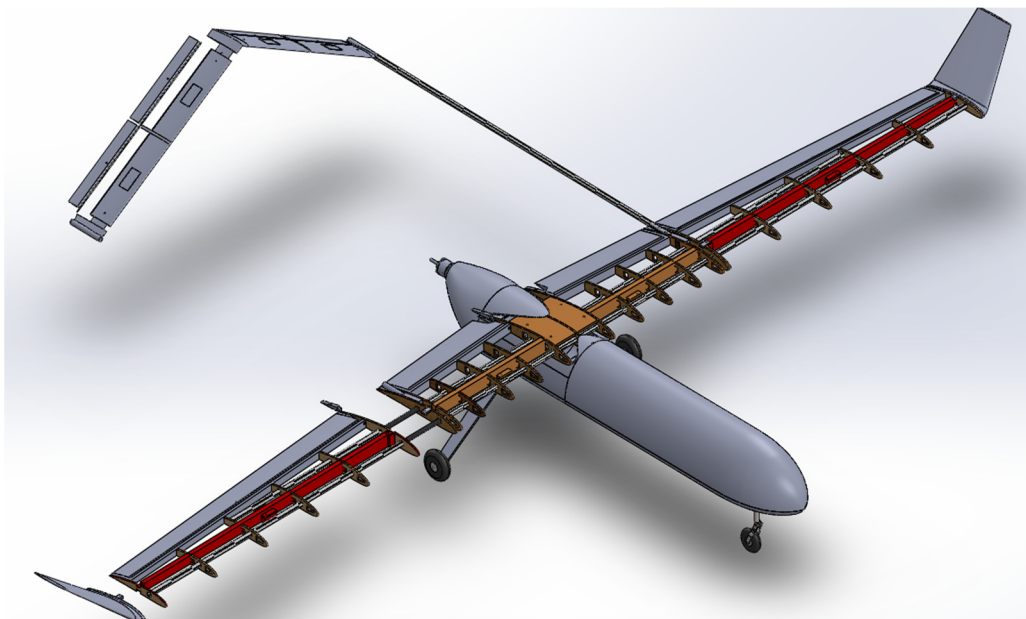


Figure 12. Real-time UAV design.

In addition, special attention is given to the landing system placement, in order to ensure the optimal stability of the UAV on the ground (see Figure 13). What can be noticed is that the “overturning” angle, which is a magnitude for evaluating the rollover tendency of the UAV while moving on the ground in closed turns, is 51.46° , less than the angle of 63° , which is the upper limit for such purposes. This means that the developed UAV is stable in case of closed turns when it moves on the ground.

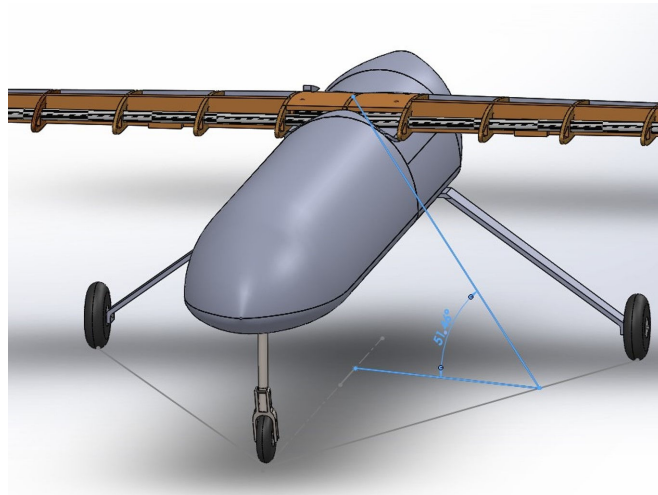


Figure 13. Positioning the landing gear of the developed UAV.

8. Construction Process

The basic criteria in the construction of the developed UAV are the choice of materials. The selected materials should minimize the weight and maximize the structural strength; at the same time, they should allow the easy processing and easy production processes [28]. Also, the construction cost and the production process should be kept as low as possible, satisfying the “cost-effective” term. In order to cover the aforementioned characteristics, the UAV components were made of fiberglass (to keep costs low compared to carbon fiber), whereas a 3D printer was used in detachable parts and other parts that are difficult to build (servo-mechanisms, control surfaces, etc.) in order to minimize the cost and construction time, while the airfoil ribs are made of plywood.

Following the above process, the construction of the UAV began with the printing of the molds on the 3D printer (see Figure 14). The fuselage was then made of fiberglass and PVC foam by using the sandwich method and the individual systems were installed (see Figure 15).

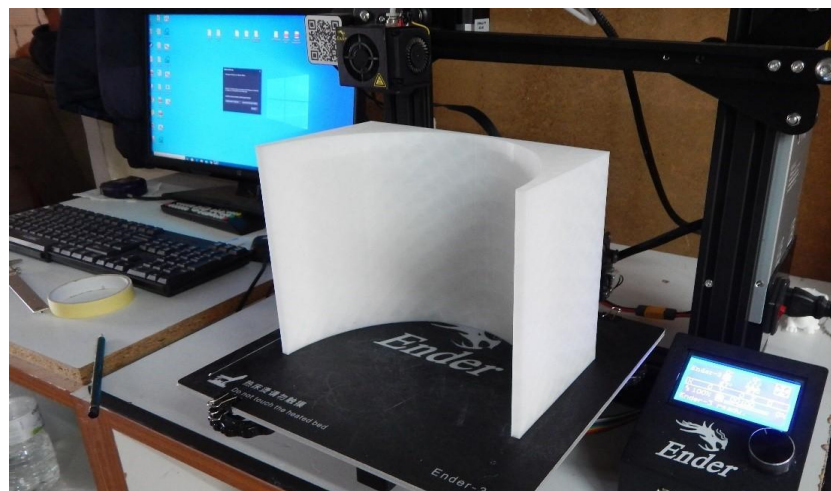


Figure 14. Making molds for the construction of the developed UAV.



Figure 15. Fuselage construction of the developed UAV.

Then, the main wing was built. Two carbon tubes were used for main spars and airfoil ribs from plywood, while the skin consists of pure fiberglass (see Figure 16).



Figure 16. Main wing construction of the developed UAV.

The tail fin was made entirely on a 3D printer, for reasons of ease of construction. That means that the structural stress it receives is low. It is also worth noting that each surface of the tail consists of two servo-mechanisms for failsafe purposes (see Figure 17).

The construction process was completed with the painting and assembly of the UAV (see Figure 18). The empty weight of the UAV is finally less than 5 kg, thus allowing for 3S 13,000 mAh LI-PO battery installation, to reach the value of the initial estimate of 5 kg empty weight.

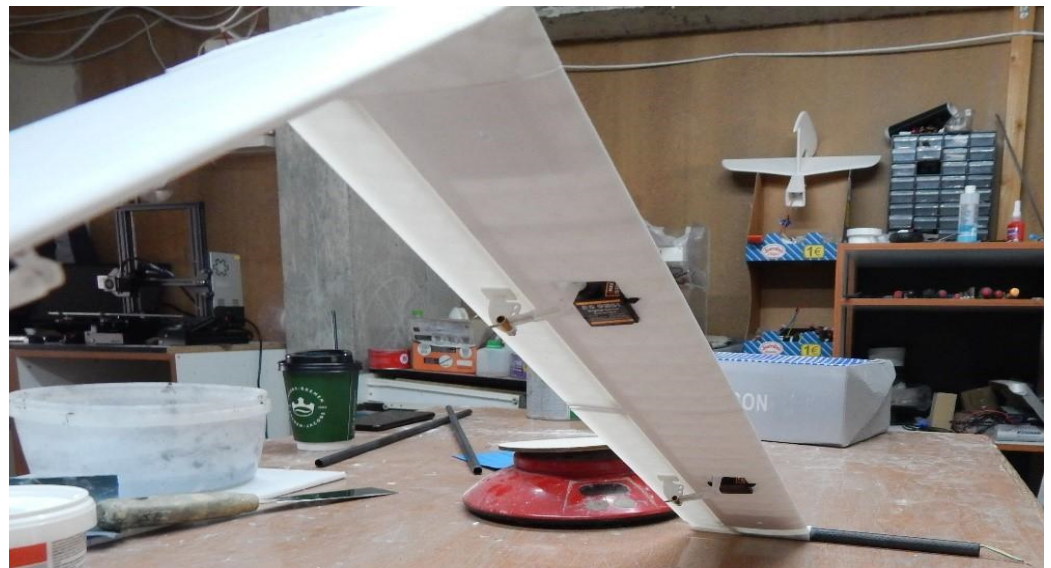


Figure 17. Tail construction of the developed UAV.



Figure 18. General fixed-wing UAV assembly.

9. Ground and Flight Testings

As mentioned previously, the experimental fixed-wing UAV consists of a propulsion system, flight control, and navigation system, battery pack, main and secondary structural modules, control surface servos, and a Ground Control Station (GCS). The Pixhawk 2.4.6 autopilot was selected for the testings as it represents low-cost autopilot software [29]. The Pixhawk 2.4.6 autopilot has an IMU module (gyroscopes and accelerometers), magnetometer (compass), barometer, GPS module, power system, and various interfaces. It can handle all the flight control and navigation requirements. Also, it has a total weight of 15.5 g, dimensions of $44 \times 84 \times 12$ mm, and operating temperature range between -40°C and 85°C . It can also offer great safety during the UAV flights, as it has a backup system for supporting all the sensors and processors. Additionally, and for failsafe purposes, the UAV has a secondary Ultimate Battery Elimination Circuit (UBEC) and two GPS modules.

9.1. Ground Testings

The ground testing refers to the tests that are performed on the ground before the flight of a UAV. In our case, the developed UAV was tested for wheel stability and ground steering quality. These tests gave information about the acceleration of the UAV due to the engine thrust and, in general, the UAV's behavior on the ground.

Two main ground tests were taken into consideration:

- The first test was carried out in manual mode, that is, without the involvement of the flight controller, in order to examine the “natural” behavior of the UAV. During the tests, a slight deviation to the left was observed, which was resolved by trimming the nose wheel. It was also observed that the wheels during the runway do not show turbulence or other undesirable features and contribute positively to the smooth acceleration of the UAV.
- The second test was carried out with the involvement of the flight controller (FBWA mode). In this case, the waving effects were observed, in which the UAV during the straight flight showed slight repetitive changes to the left and right of the path. This phenomenon can have catastrophic effects during the take-off of the UAV. After several additional tests, this problem was solved by recalculating relative values in the autopilot software.

Moreover, ground tests were executed with the addition of payload, reaching the final weight value of 6 kg. During the runway of the UAV, the static stability was confirmed, the control responses were within desirable limits, the acceleration of the engine was in desirable frames, and the contribution of the wheels was positive. The only negative point that was observed is related to the damping system of the nose wheel, which was quite weak, as it caused continuous pitch up-pitch down motions during the runway in the ground. This problem was solved by adding one damper to the nose wheel.

9.2. Flight Testings

With respect to the flight testings, manual flights were performed for safety reasons and data sharing purposes, and then semi-autonomous and autonomous flights were followed. Before the flight testings, the flight plan was reviewed carefully. When all flight preparations and inspections were completed and the weather was within acceptable limitations, the flight testings were performed on 06 January 2021, in Karditsa, Greece. The UAV had to follow a typical airfield traffic pattern, i.e., a standard path during the take-off or/and landing phases while maintaining visual contact with the airfield. Such patterns are used at airfields for air safety. Owing to the use of a consistent flight pattern, UAV pilots can know from where to expect other air traffic vehicles, thus being able to see and avoid them. When UAV pilots fly under visual flight rules (VFRs), the airfield traffic pattern is vital in keeping the UAV in safe flight mode [30].

According to the principal flight test guidelines [31], the first phase of the manual flight testings starts with the battery connection to the UAV flight controller. The UAV was left in this state for a while for the purpose of gradual heating of the flight controller, while at the same time, the last checks were made. Taking a take-off position, the final pre-flight control surfaces were inspected. This flight took place in normal mode for examining the “natural flight” of the UAV. Having the UAV already in the take-off position, the thrust stick moves to the maximum and the UAV starts accelerating. Smooth and desired acceleration was observed at this stage and the control was excellent. This was followed by a smooth ascent of the UAV with an angle at about 8°. The ascent was completed when the UAV reached an altitude of about 100 m and then a straight-level flight was followed for about 15 min. During this flight test, it was observed that the static and dynamic stability of the UAV were excellent, without the need of trimming the control surfaces, and the response to the stick was excellent. The thrust and response of the engine was also excellent for the whole flight test, but its operation amounted to 70–75% of the thrust instead of 55% (in theory). This is normal because the UAV was flying at an altitude of 100 m (due to tests) instead of 500 m (the air density was higher and therefore the drag force increased) [32,33]. Furthermore, some “extreme” manoeuvres took place, which confirmed the structural strength of the UAV. In about the middle of the flight, the flight mode switched from “normal” to “stabilize”. The flight controller greatly improved the flight characteristics of the UAV, providing additional stability and better control. The only negative was the abrupt response of the flaperons (due to the relatively large control surface area), which

was solved by reducing the response time of the corresponding servos. Finally, upon its return to the ground, the UAV was tested for structural stress without any adverse effects.

On the other hand, in the second phase of the manual flight testings, the flight mode was changed to RTL (Return-To-Land), allowing UAV to complete a repeated circle over its starting point, for about 5 min. On its return on the ground, UAV was examined for any construction failures, without the presence of negative results.

As the UAV successfully passed the above phases of the manual flight testings, the next step was the semi-autonomous flight testings. Within the frame of the semi-autonomous flight, the UAV had to take off manually, then continued the flight in “Auto” mode and completed the flight in manual landing. The main purpose of the semi-autonomous flight testings was to examine, through the flight controller, the control and the response of the UAV during the “Auto” mode. The mission was planned through the Mission Planner by selecting four waypoints in space, which are declared in terms of function (take-off, landing, speed up and down, servo trim, mode change, execution of an order, etc.) and the flight conditions (altitude, speed, etc.). The process was completed by uploading the mission to the flight controller. The flight started with the take-off of the UAV in “Stabilize” mode, and as it reached the height of 100 m, it switched to “Auto” mode. While performing the above procedure, it was observed that the UAV successfully followed the mission waypoints as the flight controller had excellent control of the UAV. Furthermore, it was observed that the response of the UAV in the roll motion was the desired one and the deviation between groundspeed (GPS-based speed) and airspeed was 1–1.5 m/s, thus confirming the airspeed calibration process.

The last phase of flight testings is the fully autonomous flight. Having confirmed the flight reliability of the UAV, all that remains for the realization of the autonomous flight is the design of its mission. The mission was planned through the Mission Planner by selecting eight waypoints in space. In general, the UAV started with the take-off phase, then it reached the selected altitude, and then a gradual descent for the landing started (Figure 19). The flight path is like a rectangle, the total distance is about 3 km, the height is up to 100 m, and at the landing stage, the glide slope is about 10%.

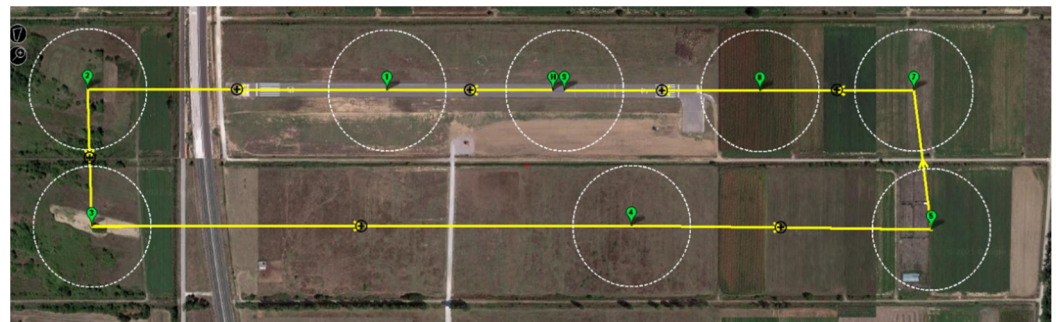


Figure 19. Autonomous flight test for the developed fixed-wing UAV.

Within the frame of the fully autonomous flight, the UAV followed the flight plan successfully. The performance of the UAV was positive in all cases with the only negative being the steep landing, where the UAV appeared to experience stalling just prior to its touch with the ground, causing an abrupt landing. In this context, numerous test flights and multiple modifications of the factors that affect the autonomous landing process took place, allowing the UAV to touch the ground smoothly.

10. Conclusions and Future Steps

This paper reports the design, manufacturing, and testing of an electrically operated flying wing UAV. The major achievements of this study include the aerodynamic performance design based on initial specifications, and the cost-effective UAV design procedure. By determining the initial specifications, all performance requirements are covered clearly.

The practical value of this study lies in the fact that it uses low-fidelity software and in-house tools to support the design and performance of the fixed-wing UAV, by also keeping low the required propulsion, low construction costs, and advanced flight autonomy.

Based on the design results, the fixed-wing UAV was built. To the best of our knowledge, this is one of the first studies where the whole main structural assembly with composite materials was performed, by using an autopilot such as Pixhawk 2.4.6. After successfully assembling all the parts together, the necessary inspections and flight plans took place. The actual total weight of 6 kg was within the initial design specifications.

In general, the main performance characteristics of the developed fixed-wing UAV are based on the following elements:

- The maximum C_L (flaperons deployed) is 1.724, and the stall speed at sea level is 10.3 m/s;
- For a payload of 1 kg and maximum take-off weight of 6 kg, the wing load is almost 120 N/m²;
- At the cruising altitude of 500 m, the cruising speed is 16 m/s, the cruising power is about 75 W, and the endurance is up to 2 h;
- The take-off distance is 24 m at a speed of 10.8 m/s;
- The maximum speed at cruising altitude is calculated as 28.8 m/s and the maximum power demand is 287.5 W;
- The absolute ceiling is calculated as 3000 m.

Furthermore, one of the main research activities that can be set up for the future is to build not only on various UAVs with specific requirements for different uses and payloads, such as food and medicine, but also in the way that UAVs will effectively and safely transport each payload depending on the shipment. Also, having already experimented enough with the capabilities of the flight software and the GCS, a future research activity could be to conduct flights in extreme weather conditions, in order to confirm both the reliability of the developed fixed-wing UAV and the capabilities of the software.

Moreover, as the selection process for the wing and tail fin airfoils of the developed fixed-wing UAV was limited to 4-digit NACA airfoils, mainly due to the simplicity of the whole design process that was followed in the present study, additional aerodynamic analysis will form part of our future research activities, to further enhance the overall design analysis and obtain even more encouraging results. Further investigations will also focus on the structural analysis of the developed fixed-wing UAV to further enhance its overall design analysis. Additionally, as the selections for the AR and angle of attack were examined in terms of the minimum drag velocity and minimum power velocity, mainly due to the simplicity of the whole design process that was followed in the present study, further investigations need to take place by considering the lift-to-drag ratio, structural analysis, aerodynamics analysis, etc.

Finally, in future missions, special emphasis will be placed on the abort landing option provided by the software, and also on the sensors that will be particularly useful in this process. Further investigations will also focus on stability and control problems of the developed fixed-wing UAV, caused by wind flow (e.g., vertical/horizontal gusts), critical scenarios in take-off and landing (e.g., sloped terrain, obstacles) phases, and pendulum effects during left/right turning, which make trajectory tracking and attitude stabilization challenging tasks.

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